

Fiodar Hancharou

☎ +48 501-677-233 ✉ correlatedpeople@gmail.com 🔗 [linkedin.com/in/fiodar-hancharou](https://www.linkedin.com/in/fiodar-hancharou) 📧 @Somesubstance

Summary

Senior AI engineer with 6+ years of experience building production-grade ML and LLM systems, certified Claude Architect. Specialized in agentic architectures, RAG pipelines, and scalable backend systems on AWS and Azure. Demonstrated success in designing and developing AI agents, recommendation systems, RAG, and API. Deep knowledge of LLM, ML algorithms, prompt engineering, evaluation, tracing, distributed systems, and databases. Strong analytical and problem-solving skills. Collaborative team player with high attention to detail. Proven track record delivering end-to-end AI solutions—from research and prototyping to deployment and monitoring—across energy, finance, and e-commerce domains.

Technical Skills

Core : Python, SQL **AI / ML:** RAG, Agentic Systems, NLP, Prompt Engineering, Evaluation, LLMs, Claude SDK, Semantic Kernel, LangGraph, Langfuse, OpenAI API, Bedrock, pgvector, PineCone, Scikit-learn, TensorFlow

Backend: FastAPI, SQLAlchemy, Pydantic, Pandas

Databases: PostgreSQL, Redis, DynamoDB, MySQL, SQLite, ElasticSearch

Infra: Kubernetes, Docker, Docker-compose, SonarQube, Git, Linux, Splunk, Bitbucket, Bamboo, Jenkins, Copilot

Clouds:

AWS - Sagemaker, Bedrock, Lambda, S3, IAM, CloudWatch

Azure - Azure Functions, DevOps, App Services, Blob storage, PostgreSQL, key vault

Work Experience

EPAM

Senior Software Engineer

Nov 2021 – Now

Wroclaw, Poland

ENERGYai AIQ | Python, Claude, OpenAI API, Semantic Kernel, FastAPI, Langfuse, LangChain, Streamlit, Databricks, PostgreSQL, OSDU, MS Fabric, Redis, MongoDB, Plotly, Pandas, GraphQL, Docker, Kubernetes, Azure

Description: Startup for boosting the performance of Oil Drilling and Probing during Site Investigation with multi-tool Agentic AI in the core

Impact Summary: Built a production multi-agent AI system for oil drilling optimization, integrating heterogeneous data sources and reducing analysis latency.

Contribution:

- * Designed a production-scale multi-agent system with tool orchestration, memory management, and retrieval integration
- * Created of a TTYD (talk to you data) retrieval systems for OSDU, Databricks, MS data Fabric
- * Handled the multimodal context (text, images, plots, tables, etc.) and storage systems for them
- * Implemented tracing evaluation via Langfuse
- * Preparing numerous demos for the client
- * Created the development-specific Streamlit UI interface for the agentic interactions

Adidas | Python, AWS CDK, RAG, PostgreSQL, Pydantic, Jenkins, Pytest, Docker, Sagemaker, lambda, Bedrock, Nodejs, Copilot

Description: IaaS (infrastructure as a code) for RAG-based system used for efficient information retrieval from a Confluence knowledge base.

Contribution:

- * Defining the project architecture
- * Creating constructs for definition of CloudFormation stacks, networks, VPCs, subnets, Postgres DB, prediction lambda, IAM roles, S3 bucket for statistics, secret manager, event bridge, Sagemaker pipeline resources
- * Writing and debugging the Jenkins scripts for CI/CD
- * Providing a knowledge transfer session to colleagues

Vanguard | *Python, Pydantic, pytest, Postman, Docker, AWS, Splunk, Redis, DynamoDB, GateKeeper*

Description: P2P Transactions functionality of a trading platform. The project consists of AWS lambdas that handle validations, information retrieval, and financial operations logic.

Contribution:

- * Design and implementation of MemCache for auth tokens in Redis.
- * Refactoring the services utilizing the "adapter" pattern. As a result, the repeatable code was reduced by 70%, several data type mismatches were identified, and unnecessary cyclomatic complexity was avoided.
- * Logging system refactoring. Unnecessary logs were removed, and "excepthook" was implemented to store the full traceback of critical bugs.
- * Creation of a library for legacy code decomposition, which reduced repeatable code and simplified further development.
- * people onboarding and knowledge transfer

Shell | *PostgreSQL, Python, FastAPI, SQLAlchemy, Pandas, Pydantic, Docker, Alembic, Azure*

Description: PERC (probabilistic emission reduction calculator) – the forecasting system for carbon credit generation projects. The core functionality is represented by a forecast calculation of an annual emission reduction for a mangrove ecosystem.

Contribution:

- * Analyzed and refactored the core Monte Carlo simulation code, improving its performance and readability, utilizing Multiprocessing.
- * Successfully pitched to the team and business the "edit" menu design change towards a more user-friendly one
- * Found several critical logic bugs left from the POC stage
- * Developed the entire back-end part, database layer using SQLAlchemy and FastAPI

Shell | *Python, Pandas, Sklearn, nltk, Azure.functions, Azure MySQL, pandas, TF-IDF*

Role: Data Scientist, Backend Developer

Description: The recommendation system for the e-commerce platform

Contribution:

- * Conducted full design cycle of development from customer consultation and business requirements clarification to POC and MVP
- * Created Time-triggered Azure function for data retrieval, text data transformations, cosine similarities matrix computation, and saving recommendations to the database
- * Created an HTTP-triggered Azure function for retrieving recommendations for specific products

Inria

Nov 2019 - Jul 2021

Software Engineer

Lille, France

Project 1 | *Python, Flask, SQLAlchemy, Pandas, OpenCV, Kubernetes, Docker, MySQL, Linux, HTML, CSS, Bootstrap, Pytest*

Role: Full Stack Software Engineer

Description: The IoT application that provides remote control, analysis, and online video monitoring for the Inverted Pendulum in the laboratory. The platform handles the stream of measurements from the pendulum sensors, evaluates the control quality metrics, stores the results, and provides easy remote access to users.

Contribution:

- * Communications with customer
- * Prototyped custom implementation for Message Queue using FTP service for data exchange in the local network
- * Designed and implemented the entire app UI, FE, BE
- * Design of data models and interactions with the database
- * Design and implementation of UI
- * Organized CI/CD system using Docker, GitHub actions, and a Linux system on Raspberry Pi
- * Troubleshooting and fixing issues
- * Production monitoring and support

Publications

Title: Deep Learning for Power System Security Assessment

Novelty: For the first time, a method for the representation of power system states in the form of images was introduced. The dataset of such images was used to train and validate the convolutional neural network that classifies power system states in terms of transient stability

Details: The paper presented at 13th IEEE PES PowerTech 2019 Conference, Milan. Published, Scopus indexed

Title: Testing machine learning approaches for wind plants power output prediction

Novelty: Improvement of wind plant's power output predictions using the data-driven models based on historical time series data of power output and weather forecasts. The scores from different models were compared with each other and with a conventional physical model

Details: The paper presented at the IEEE REEPE 2019 conference, Moscow. Published, Scopus indexed

Education

Skolkovo Institute of Science and Technology

2017-2019

Master of Science in Information Systems and Technology (GPA: 4.6 / 5)

Relevant Coursework: Machine Learning, Deep Learning, Math Optimization (Python, Tensorflow, numpy, pandas)

Thesis Title: Machine Learning Methods for Power System Security Assessment

Belarusian National Technical University

2012-2017

Power Engineering (GPA: 8.9 / 10)

Relevant Coursework: Informatics (Fortran, Matlab), Mathematics (Linear Algebra, Probability Theory)